

Computational Fluid Dynamics Modeling of Combustion of Synthetic Fuel of Thermochemical Heat Recuperation Systems

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Abstract: CFD modeling of the combustion of synthetic fuel formed in the systems of thermochemical recuperation of waste flue gas heat due to steam methane reforming was performed using the ANSYS Fluent software. Scientific justification and validation of the physicomathematical approaches involved the ANSYS Fluent for the problems of modeling the combustion of multicomponent hydrogen-containing gas mixtures. Numerical results were validated against experimental data. A visual comparison of the flame contours obtained by burning syngas at Reynolds numbers of 600, 800, and 1000 was performed. In all cases there is obvious convergence of the results. Change in the temperature of the fuel–air mixture at the entrance to the combustion chamber was found to have no significant effect on the temperature of the combustion products. The obtained results are of practical importance for the design of burner units of high-temperature plants with thermochemical heat recuperation.

Keywords: hydrogen, combustion, syngas, thermochemical recuperation, CFD modeling.

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INTRODUCTION

The energy and environmental problems facing modern society have motivated the continuous improvement of power plants, the search for new fuels, and the development of innovative technologies for the use of traditional hydrocarbon fuels. In recent decades, experts and engineers from different countries have paid great attention to the use of hydrogen as a fuel source of energy. Thus, according to the Hydrogen Scenario of the International Energy Agency (IEA), large-scale hydrogen production on a global scale will increase by 20–25% annually after 2030 [1]. The main consumers of hydrogen should be transport and power plants using this gas as fuel. Known problems of storage and transportation of hydrogen to consumers determine the use of technologies in a transitional stage in which fuel hydrogen is produced directly by a fuel-consuming unit [2].

Production of hydrogen-containing gas immediately before consumption in a power plant has found application in high-temperature plants with thermochemical recuperation of exhaust flue gas heat [2–5]. The essence of this process is that the physical heat of waste flue gases is used for preliminary endothermic processing of the initial fuel, which is thus supplied with an additional amount of chemically bound heat. One way to implement this process is its recuperation due to steam methane reforming. The heat of the power plant flue gas is used for surface heating of the steam reforming reactor in which chemical reactions occur to produce syngas (hydrogen-containing fuel).

In high-temperature plants with thermochemical heat recuperation due to steam methane reforming, hydrogen-containing gas is used as fuel by the same plant [6]. A distinctive feature of the use of syngas as fuel compared to natural gas is the different combustion characteristics of the fuels. Combustion of syngas of different compositions is well studied both experimentally and theoretically. However, the literature does not cover the whole range of practically interesting is-

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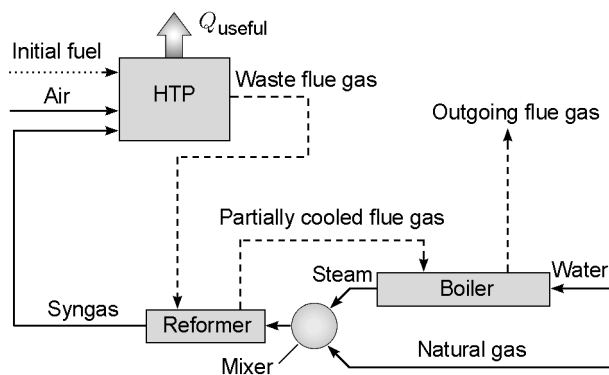


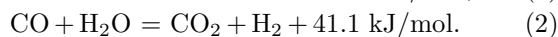
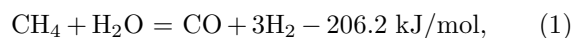
Fig. 1. High-temperature plant with thermochemical heat recuperation [11].

sues, in particular, the combustion of high-temperature hydrogen-containing gases in high-temperature plants with thermochemical heat recuperation.

The development of numerical methods for studying heat and mass transfer processes on high-performance computers has opened new prospects for the study of the combustion of fuels of various composition under various initial conditions. For numerical modeling [computational fluid dynamics (CFD) model] of combustion processes, specialized software products such as ANSYS (Fluent and CFX), Comsol Multiphysics, OpenFOAM, etc., are used [7–9]. Numerical modeling of the combustion of gaseous fuel includes the calculation of processes such as heat transfer, mass transfer, diffusion, etc. [10].

This paper deals with a numerical investigation of the combustion characteristics of syngas produced in a thermochemical heat recuperation system. The study was performed using the ANSYS Fluent (Academic Research) software product.

A schematic diagram of a high-temperature plant (HTP) with thermochemical recuperation of exhaust flue gas heat due to steam methane reforming is shown in Fig. 1 [11]. Exhaust flue gases of the HTP are sent for surface heating of the reaction space of the reformer in which the highly endothermic reaction of steam methane reforming takes place:



The reforming product [syngas having a higher heat of combustion (per 1 mole of the initial fuel) than the initial methane]—is sent for combustion to the combustion chamber of the HTP, to which blowing air is also fed. Steam produced using the residual heat of flue gases after the reformer is mixed in the gas mixer with the initial fuel—natural gas, whose main component is methane.

Table

$T, \text{ K}$	Equilibrium composition of syngas, vol. %				
	H_2	CO	H_2O	CO_2	CH_4
700	33.04	4.07	32.74	6.52	23.63
800	58.11	9.86	18.94	7.23	5.86
900	61.34	12.89	16.51	6.08	3.18
1 000	63.40	14.38	16.12	5.08	1.02
1 100	63.65	16.14	16.15	3.98	0.08

The molar concentration of each component of syngas at the outlet of the thermochemical reformer depends on the technological parameters of the thermochemical heat recuperation system (temperature, composition of the initial reaction mixture, pressure) as well as on the design characteristics of the reformer (volume of the reaction space, type and amount of catalyst, and the velocity of the reaction mixture) [3, 11]. Steam methane reforming has been well studied both experimentally and theoretically. Modern designs of steam reforming reactors provide syngas composition as close as possible to the equilibrium one [3, 11]. The equilibrium syngas composition obtained by thermodynamic analysis of steam methane reforming using the IVTANTERMO software simulator developed by specialists of the Joint Institute for High Temperatures of the Russian Academy of Sciences is given in the table. The initial data for the thermodynamic analysis are a system pressure of 0.1 MPa and a reaction mixture of composition $\text{H}_2\text{O} : \text{CH}_4 = 2 : 1$.

The numerical study of the combustion characteristics of the hydrogen-containing gas of the composition shown in the table was carried out for burner conditions with forced air supply and complete premixing of the fuel and air. The computational domain and computational grid are shown in Fig. 2a. The problem was solved in a two-dimensional axisymmetric formulation. To improve the quality of the solution, the computational grid was adapted to solve the problem as shown in Fig. 2b. The computational domain includes a inlet pipe 8 mm in diameter and 40 mm long and a combustion chamber 50 mm in diameter and 100 mm long. The centerline is the x axis, and the radial direction is the y axis. The coordinate origin along the x axis lies on the exit line of the inlet pipe to the combustion region. The length of the inlet pipe was selected so as to provide developed flow of the gas mixture at the outlet. The diameter of the channel in the combustion region was selected so that the walls did not have a significant effect on the flame structure. Studies were carried out for different Reynolds numbers ($\text{Re} = 600\text{--}1000$).

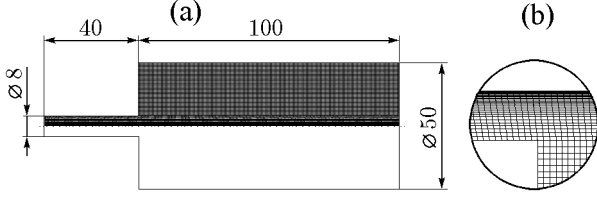


Fig. 2. (a) Computational domain and computational grid; (b) adaptation of the computational grid for the formulated problem; dimensions are in millimeters.

In the present work, the hydrogen-containing mixture burns in air at a stoichiometric ratio (fuel/air equivalence ratio $\alpha = 1.0$). To solve the formulated problem, the continuity, momentum, and energy equations and the equations of chemical combustion kinetics of the hydrogen-containing gas were jointly solved [12, 13]. In the modeling, we used the following assumptions that do not have a significant effect on the final result: the flow of the combustible gas mixture is stationary, the effect of viscosity is not taken into account, and surface chemical oxidation of the burner components do not occur.

The continuity equation written in general form for the stationary flow has the form

$$\nabla(\rho \mathbf{u}) = 0, \quad (3)$$

where $\mathbf{u}$ is the velocity of the reaction mixture and ρ is the density of the mixture.

The equation of conservation of momenta is given by

$$\nabla(\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla(\bar{\tau} + \tau') + \rho \mathbf{g} + \mathbf{F}, \quad (4)$$

where p is the static pressure, $\rho \mathbf{g}$ and $\mathbf{F}$ are the gravitational and external forces, respectively, and τ' is the Reynolds stress tensor. The stress tensor $\bar{\tau}$ in Eq. (4) is given by the formula

$$\bar{\tau} = \mu \left[(\nabla \mathbf{u} + (\nabla \mathbf{u})^T) - \frac{2}{3} \nabla \mathbf{u} I \right], \quad (5)$$

where μ is the molecular viscosity and I is the unit tensor. The first term of expression (5) is the strain rate tensor, and the second term takes into account the volumetric expansion.

The energy conservation equation for the flow area of the combustion chamber is given by

$$\begin{aligned} \nabla \mathbf{u} (p + \rho E_f) = \nabla \left[\lambda_{\text{eff}} \nabla T - \left(\sum h_i \mathbf{J}_i \right) + \bar{\tau} \mathbf{u} \right] \\ + \sum h_i R_i - \nabla q_r, \end{aligned} \quad (6)$$

where E_f is the total energy of combustion of the components, $\mathbf{J}_i$ is the diffusion flow of the i th component,

λ_{eff} is the thermal conductivity, R_i is the rate of the i th reaction, h_i is the enthalpy of formation or consumption of the j th component, and q_r is the heat transferred to the wall by radiation; the subscript i refers to the chemical reactions and the subscript j to the component involved in the i th reaction.

The transfer equation for the combustion components and products is written as

$$\nabla(\rho \mathbf{u} g_i) = -\nabla \mathbf{J}_i + R_i, \quad (7)$$

where R_i is the rate of formation or consumption of the i th component.

The diffusion flow is

$$\mathbf{J}_i = - \left(\rho D_{m,i} + \frac{\mu_t}{Sc_t} \right) \nabla g_i - D_{T,i} \frac{\nabla T}{T}, \quad (8)$$

where $D_{m,i}$ and $D_{T,i}$ are the coefficients of mass and thermal diffusion of the i th substance, respectively, Sc_t is the Schmidt number, g_i is the mass fraction of the i th component, and T is the temperature.

The combustion of syngas was modeled using a system of three global chemical reactions proposed in the Species Transport module of the Fluent solver: $\text{CH}_4 + \text{O}_2 = \text{CO}_2 + 2\text{H}_2\text{O}$, $\text{CO} + 0.5\text{O}_2 = \text{CO}_2$, and $2\text{H}_2 + \text{O}_2 = 2\text{H}_2\text{O}$. In the calculations, the vortex dissipation model based on the following assumptions was used: the chemical reaction is fairly fast relative to the diffusion processes in the reacting flow; mixing of the reagents at the molecular level leads to the instantaneous formation of reaction products; the reaction rate can be directly related to the time required for the reagents to be mixed at the molecular level. It is known that these assumptions are valid in the case where the chemical reaction rate is many times higher than the rate of other physical processes, in particular, diffusion.

As can be seen from the table, the main combustible component of syngas is hydrogen, whose combustion is characterized by high Damköhler number ($Da \gg 1$) [9]. In addition, the syngas produced at a temperature of 700 K contains a noticeable amount of methane, whose combustion reaction is also characterized by a high rate. However, the syngas includes carbon monoxide CO, whose burning rate is significantly lower than the burning rate of methane and hydrogen. In this connection, we determined the Damköhler number for the combustion reaction of carbon monoxide ($\text{CO} + 0.5\text{O}_2 = \text{CO}_2$) for which the rate was determined according to the expression [14]

$$\frac{d[\text{CO}]}{dt} = 2.24 \cdot 10^{12} \exp \left(-\frac{1.7 \cdot 10^8}{RT} \right) [\text{CO}][\text{O}_2]^{0.25},$$

where $2.24 \cdot 10^{12}$ is the pre-exponential factor (in s^{-1}), $1.7 \cdot 10^8$ is the activation energy (in J/mol),

[CO] and [O₂] are the mole fractions of carbon monoxide and oxygen, respectively, and R [J/(mol·K)] is the universal gas constant.

It is seen from expression (9) that the rate of the combustion reaction increases with increasing temperature. Thus, at $T = 700$ K, the Damköhler number $Da > 5.0$, which also increases with increasing temperature. In this regard, we can assume that the use of the eddy dissipation models (EDM) is justified for the given initial conditions—high temperature of the fuel mixture and low mole fraction of CO.

Advantages of the EDM over more computationally costly models (for example, PDF with the mechanism of chemical reactions of combustion) are relatively low cost of computing power with a sufficiently accurate modeling of the combustion process at $Re > 500$. In the case of using the vortex dissipation model, the stagewise nature of combustion reactions is not taken into account, and the rate of global reactions given above is determined by the time scale of turbulent mixing of reagents. Using the EDM in the Fluent solver is only possible for various types of the k - ε model turbulence. In this connection, the time-averaged Navier–Stokes equations in the adopted model are closed by the standard k - ε turbulence model, which allows a fairly low-cost solution of problems in a wide range of Reynolds numbers. The kinetic energy (k) and the rate of dissipation of turbulent energy (ε) can be determined from the following expressions for stationary flow:

$$\nabla(\rho k \mathbf{u}) = \nabla \left[\left(\mu + \frac{\mu_t}{Pr_k} \right) \nabla k \right] + G_k + G_b - \rho \varepsilon - Y_M + S_k; \quad (10)$$

$$\nabla(\rho \varepsilon \mathbf{u}) = \nabla \left[\left(\mu + \frac{\mu_t}{Pr_\varepsilon} \right) \nabla \varepsilon \right] + C_{1\varepsilon} \frac{\varepsilon}{k} (G_k + C_{3\varepsilon} G_b) - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} + S_\varepsilon. \quad (11)$$

Here μ_t is the turbulent viscosity, Pr is the turbulent Prandtl number, $C_{1\varepsilon}$, $C_{2\varepsilon}$, and $C_{3\varepsilon}$ are constants of the k - ε turbulence model, G_k is the generation term, G_b is the kinetic energy of the buoyancy force, and Y_M is the contribution of the variable expansion with compression turbulence to the total dissipation rate (usually taken into account at high Mach numbers). In Eqs. (10) and (11), the following values of constants are used: $C_{1\varepsilon} = 1.44$, $C_{2\varepsilon} = 1.92$, $C_{3\varepsilon} = 1$, $Pr_k = 1.0$, and $Pr_\varepsilon = 1.3$.

The boundary and initial conditions for the computational domain shown in Fig. 2 are as follows: In the exit section of the air–fuel mixture from the supply channel (at the entrance to the combustion region),

$$x = 0, \quad T_g = T_{g(\text{ent})}, \quad g_i = g_{i(\text{ent})}; \quad (12)$$

In the exit section from the combustion region,

$$x = L, \quad \frac{\partial g_i}{\partial x} = 0; \quad (13)$$

On the centerline of the combustion region,

$$y = 0, \quad \frac{\partial g_i}{\partial y} = 0, \quad \frac{\partial T_g}{\partial y} = 0; \quad (14)$$

On the wall,

$$\frac{\lambda_w \partial^2 T}{\partial x^2} = 0, \quad \frac{\lambda_w \partial^2 T}{\partial y^2} = 0. \quad (15)$$

Since the temperature of the flame front is greater than 2500 K and since the gas flow contains three-atom components, such as H₂O and CO₂, and methane, radiant heat transfer in the combustion region will affect the enthalpy loss in the gas flow. Although some authors believe that this effect during hydrogen combustion is insignificant [15]; we use the P-1 radiation model, which is a simplified version of the more general P-N model based on the decomposition of the radiation intensity in an orthogonal series of spherical harmonics [9, 15].

In the P-1 model, radiant heat transfer is considered as a diffusion process, with the following assumptions taken into account: local thermodynamic equilibrium in the medium; the radiation field is uniform in space, and there is no pronounced radiation direction. The radiant heat flux is defined as

$$q_j^r = -\Gamma \frac{\partial G}{\partial x_i}, \quad (16)$$

where G is the integral of the radiation intensity and Γ is the effective transfer coefficient, which, under the above assumptions, has the form

$$\Gamma = 1/3\kappa, \quad (17)$$

where κ is the dispersion coefficient.

The differential transport equation for the integral of the radiation intensity is written as [16]

$$\frac{\partial}{\partial x_i} \left(\frac{1}{\kappa} \frac{\partial G}{\partial x_i} \right) + \kappa(4\sigma T^4 - G) = 0, \quad (18)$$

where σ is the Stefan–Boltzmann constant.

In the calculations, the convergence criterion was set at a level of 10^{-7} for the mass fraction of the combustion components, the equations of momentum and the continuity equation, and at a level of 10^{-4} for the convergence of the energy conservation equation [17]. Calculations were made on a computing cluster with processors based on Intel Xeon E5-2640. Convergence of the solution for most calculations was achieved in the range of iterations of 480–510.

The model was validated against experimental data on the combustion of a fuel mixture of 68% CH₄, 22% CO₂, and 10% H₂ with complete premixing at

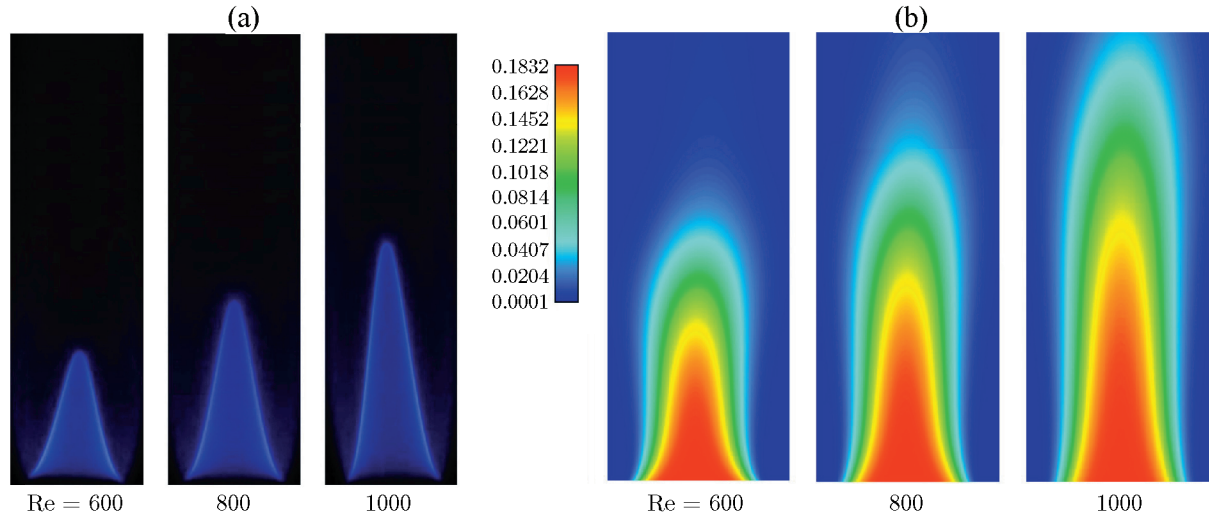


Fig. 3. Comparison of the contours of the mass fraction O_2 ($T = 300$ K): (a) experiments [13]; (b) calculation results.

$\alpha = 1.0$ [13]. Figure 3 shows the experimental results and the contours of the calculated mass fraction of oxygen at various Reynolds numbers. The kinematic viscosity of the gas mixture for determining the Reynolds numbers is calculated according to the Sutherland–Tisel rule. Visual comparison of the flames shows good convergence of the results obtained using the model and the experimental data. In the modeling results, the characteristic Bunsen cone (the combustion front) is seen.

Visual comparison of the modeling and experiment results does not allow us to ascertain the adequacy of the selected settings of the Fluent solver and hence the differential equations describing the processes. For additional verification of the results of the model, the temperature profiles of the cylindrical wall of the combustion region obtained by computational experiment were compared with the results of a physical experiment. The initial data were the results of experimental studies [15] of hydrogen combustion in a combustion chamber of small size: 100% hydrogen as fuel, hydrogen consumption of $5.25 \cdot 10^{-7}$ kg/s, and $\alpha = 1.0$.

Figure 4 shows the results obtained using the present mathematical model implemented in ANSYS Fluent and the results of experiment [15] and numerical study [16]. The relative distance was determined as the ratio of the axial distance z to the total length of the combustion chamber L . Here $z/L = 0$ corresponds to the exit of the air–fuel mixture from the inlet pipe and $z/L = 1$ to the exit of the combustion products from the combustion chamber. Figure 4 demonstrates good convergence between the modeling results and experimental data. Maximum differences in wall temperature T_w are observed near the exit of the air–

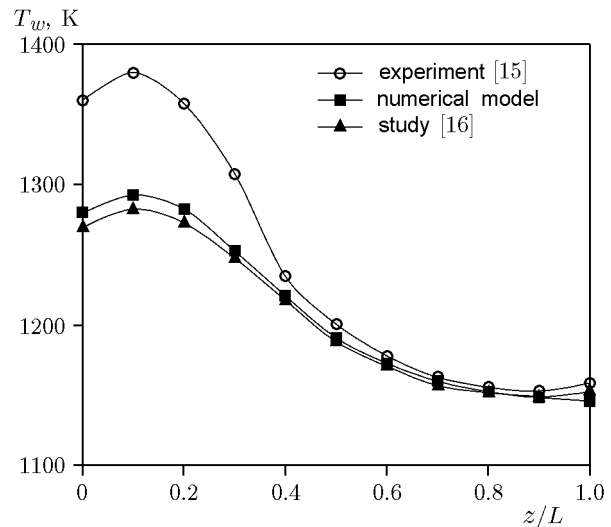


Fig. 4. Temperature distribution of the wall along the length of the combustion chamber at a hydrogen consumption of $5.25 \cdot 10^{-7}$ kg/s at $T = 300$ K.

fuel mixture from the inlet pipe. They can be explained by the instrumental error of the instruments, measurements and the assumptions made in the mathematical model. However, the standard deviation for the entire computational domain presented in Fig. 4 does not exceed 0.036.

The results of validation of the model against experimental data in Figs. 3 and 4 suggest the correctness of the selected Fluent solver settings.

In numerical CFD modeling, the accuracy of results is determined not only by the selected solver settings and convergence criteria, but also by the number of unit

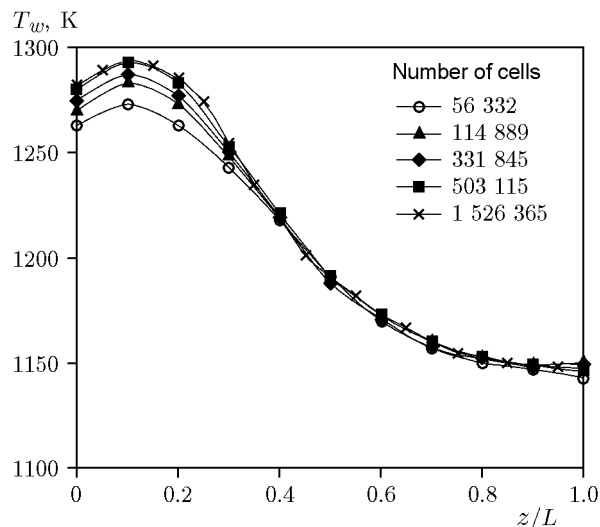


Fig. 5. Wall temperature distribution along the length of the combustion chamber for different numbers of cells for $Re = 1000$ and $T = 300$ K.

cells and the quality of the computational grid [18]. In this study, the minimum number of cells for the computational domain shown in Fig. 2 was evaluated. The following initial data were used: 100% hydrogen as fuel, a hydrogen consumption of $5.25 \cdot 10^{-7}$ kg/s, and $\alpha = 1.0$. The number of cells ranged from 56 332 to 1 526 365 with the constant convergence criteria mentioned above.

From Fig. 5, it is seen that the calculation accuracy increases with increasing number of cells; however, for a number of cells more than 500 000, the results practically do not change. Therefore, for further calculations, it is recommended to choose the number of cells in the range 500 000 to 600 000. In this case, there is a satisfactory quality of the results with minimum consumed computing power.

Figure 6 shows the temperature distribution along the axis of the flow region of the combustion chambers for syngas of various compositions and temperatures (see the table). The number of computational cells is 503 115. The calculations were performed for $\alpha = 1.0$ and a syngas flow rate of $1.35 \cdot 10^{-4}$ kg/s. The temperature of the fuel-air mixture was considered equal to the temperature at which the syngas according to the table was obtained. The obtained dependences suggest that the composition and temperature of the syngas studied do not have a significant effect on the temperature of the combustion products at the exit of the combustion chamber. This can be attributed to the effect of dissociation of water steam and carbon dioxide molecules in the combustion region adjacent to the flame front and to radiant heat exchange between the combustion products and the wall of the combustion chamber.

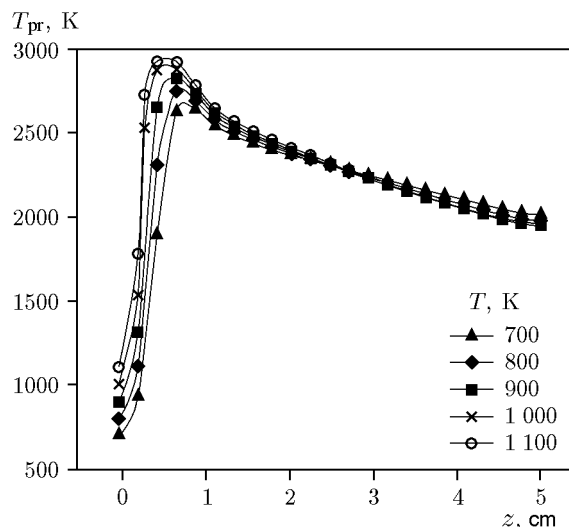


Fig. 6. Temperature distribution of combustion products along the axis of the combustion region of syngas of different compositions for $Re = 1000$.

It is known that the dissociation of water and carbon dioxide molecules is accompanied by absorption of the heat resulting from fuel combustion. Here we can consider both thermal dissociation and the dissociation of chemical compounds. For example, the multistage nature of the reactions by the chain-thermal mechanism leads to a non-equilibrium effect of the chemical dissociation of the reacting substances with the formation of intermediate radicals (e.g., the water dissociation reaction $H_2O = OH^* + H^*$). Due to this, there is heat absorption from the reaction zone and, as consequence, a decrease in temperature. This effect is observed near the flame front at high temperatures. However, as the combustion products move away from the flame front, their composition approaches the equilibrium one, and at a considerable distance from the front, it becomes the equilibrium one. In this case, chemical dissociation does not have a significant effect on the temperature of combustion products.

To quantify the effect of radiant heat transfer on the temperature of products combustion, in the settings of the Fluent solver the P-1 radiation model was disabled. Moreover, all other parameters of the mathematical model, such as the structure of the computational grid, initial conditions, turbulence model, approximation scheme, etc., remained identical. Figure 7 shows the temperature distribution of combustion products T_{pr} along the axis of the combustion chamber without accounting for radiant heat transfer. It is seen that in the combustion region adjacent to the flame front (0.8–1.2 cm), the decrease in the temperature of combustion products is stronger than that behind the flame

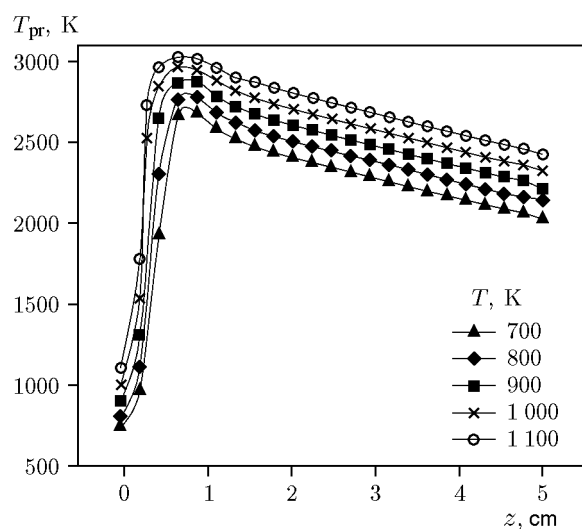


Fig. 7. Temperature distribution of combustion products along the axis of the combustion region of syngas of different composition for $Re = 1000$ without accounting for radiant heat transfer.

front. The smooth monotonic reduction in axial temperature behind the flame front is due to mixing of combustion products near the wall.

The obtained dependences suggest that the temperature of the products in the combustion region adjacent to the flame front has an effect on heat transfer by radiation as well on the dissociation of water and carbon dioxide molecules. Behind the flame front, the effect of dissociation becomes negligible and radiant heat transfer has a marked effect. Obviously, in the case $T_{in} = 1100$ K, the enthalpy loss is maximal. This makes it possible to explain the similarity of the temperature values at the outlet for all investigated values of the initial temperature (see Fig. 6).

The curves in Fig. 6 can be used to design combustion chambers of high-temperature plants with thermochemical heat recuperation. Also, the results obtained show that when converting a HTP to thermochemical heat recuperation, it is necessary to take into account the change in the temperature mode of operation of the plant. The temperature in the working chamber can be controlled by recirculation of flue gases after the steam boiler (see Fig. 1).

Figure 8 shows the temperature and velocity contours for combustion of syngas obtained at a temperature of methane reforming of 700 K and an air equivalence ratio $\alpha = 1.0$. These contours provide a visual representation of the combustion processes of the fuel–air mixture.

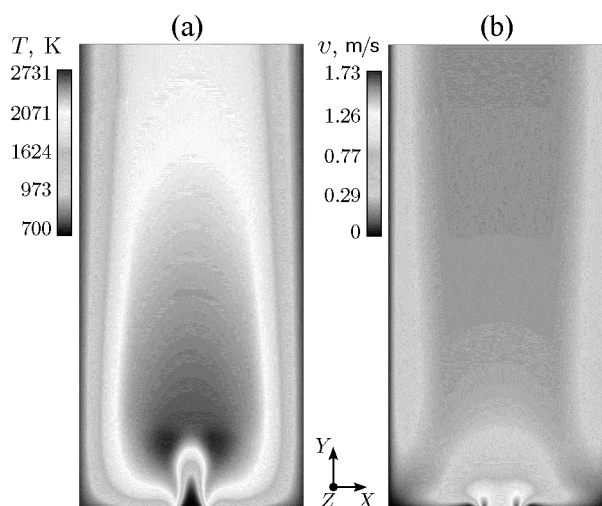


Fig. 8. Contours of temperature (a) and flow velocity (b) for $Re = 1000$ and $T_{in} = 700$ K.

The CFD modeling of the combustion of a gaseous fuel composed of several combustible components such as hydrogen, carbon monoxide, and methane, allowed us to determine the thermal characteristics of the process at each point of the combustion chamber. The model uses well-known physicomathematical models embedded in the ANSYS Fluent software code. Justification and validation of the approaches used in ANSYS Fluent to model the combustion of multicomponent hydrogen-containing gas mixtures were performed. Using the model, it was established that increasing the temperature of the fuel–air mixture before burning does not lead to a significant change in the temperature mode in the combustion chamber. For all fuel compositions studied, the temperature of the products at the exit of the combustion chamber is about 2000 K. There is good convergence between the numerical results and experimental data. The model was validated against experimental data for combustion of a fuel mixture of $68\% \text{CH}_4 + 22\% \text{CO}_2 + 10\% \text{H}_2$ with complete pre-mixing at $\alpha = 1.0$. Visual comparison of physical and numerical flames shows good convergence between the results obtained using the model and experimental data. In the modeling results, the characteristic Bunsen cone (the combustion burning front) is seen.

It has been established that increasing the number of unit cells in the computational domain over 500 000–600 000 does not lead to a noticeable increase in the calculation accuracy. The model was implemented in the software product ANSYS Fluent. The obtained data can be used in the design of burner units of high-temperature plants with thermochemical recuperation of flue gas heat due to steam methane reforming.

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